Problem Set 62

Derivatives 1

April 7, 2026

Problem 1

Find $f'(a)$ using the limit process:

(a) $f(x) = \dfrac{2x+1}{x+3}$

$$\begin{aligned}
f(x) =& \frac{2x+1}{x+3} = \frac{-5}{x+3} + 2 \\
f'(x) =& \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
=& \lim_{h\to 0} \frac{\frac{-5}{x+h+3} + 2 - \frac{-5}{x+3} - 2}{h} \\
=& \lim_{h\to 0} \frac{\frac{5}{x+3} - \frac{5}{x+h+3}}{h} \\
=& \lim_{h\to 0} \frac{\frac{5h}{(x+3)(x+h+3)}}{h} \\
=& \lim_{h\to 0} \frac{5}{(x+3)(x+h+3)} \\
=& \boxed{\frac{5}{x^3+6x+9}}
\end{aligned}$$

Problem 4

Let $f(x) = \dfrac{(2^x + 2^{-x})\sin x\sqrt{\arctan(x^2 - x + 1)}}{(7x^2+3x+1)^3}$. Use the limit definition of the derivative to find the value of $f'(0)$.

$$\begin{aligned}
f'(0) =& \lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} \\
=& \lim_{x\to 0} \frac{\frac{(2^x+2^{-x})\sin x\sqrt{\arctan(x^2-x+1)}}{(7x^2+3x+1)^3} - \frac{(2^0+2^{-0})\sin 0\sqrt{\arctan(0^2-0+1)}}{(7\cdot 0^2+3\cdot 0+1)^3}}{x} \\
=& \lim_{x\to 0} \frac{\frac{(2^x+2^{-x})\sin x\sqrt{\arctan(x^2-x+1)}}{(7x^2+3x+1)^3}}{x} \\
=& \lim_{x\to 0} \frac{(2^x + 2^{-x})\sqrt{\arctan(x^2 - x + 1)}}{(7x^2+3x+1)^3} \cdot \frac{\sin x}{x} \\
=& \lim_{x\to 0} \frac{(1+1)\sqrt{\arctan 1}}{1^3}
\end{aligned}$$

$$=2\sqrt{\frac{\pi}{4}}=\boxed{\sqrt{\pi}}$$

Theorem 1 Differentiability implies Continuity

If f is differentiable at a, then f is continuous at a.

Note: The converse of Theorem 1 is false; that is, there are functions that are continuous but not differentiable.

Problem 5

Prove Theorem 1.

Given that $f(x)$ is differentiable at a,

$$\begin{aligned}\lim_{x\to a} f(x)-f(a) &= \lim_{x\to a}(f(x)-f(a))\\ &= \lim_{x\to a}\left(\frac{(f(x)-f(a))(x-a)}{x-a}\right)\\ &= \lim_{x\to a}\left(\frac{f(x)-f(a)}{x-a}\right)\lim_{x\to a}(x-a)\\ &= f'(a)\cdot 0\\ &= 0\end{aligned}$$

$$\boxed{\lim_{x\to a} f(x)=f(a)}$$

Therefore, $f(x)$ is continuous on a.

Problem 7

Discuss the differentiability of

(b) $f(x)=\begin{cases}\dfrac{\sin x^2}{x} & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$ at $x=0$

$$f'_-(0)=\lim_{h\to 0^-}\frac{\frac{\sin h^2}{h}-0}{h}=\lim_{h\to 0^-}\frac{\sin h^2}{h^2}=1$$

$$f'_+(0)=\lim_{h\to 0^+}\frac{\frac{\sin h^2}{h}-0}{h}=\lim_{h\to 0^+}\frac{\sin h^2}{h^2}=1$$

$$f'_-(0)=f'_+(0)=1 \implies f'(0)=1$$

$$\boxed{f(x) \text{ is differentiable at } x=0}$$

(c) $f(x)=\begin{cases}x\sin(\ln x^2) & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$ at $x=0$

$$f'_-(0)=\lim_{h\to 0^-}\frac{h\sin(\ln h^2)-0}{h}=\lim_{h\to 0^-}\sin(\ln h^2)\,\text{DNE}$$

$$f'_+(0) = \lim_{h\to 0^+} \frac{h\sin(\ln h^2) - 0}{h} = \lim_{h\to 0^+} \sin(\ln h^2) \text{ DNE}$$
$$f'_-(0) \text{ DNE}, f'_+(0) \text{ DNE} \implies f'(0) \text{ DNE}$$
$$\boxed{f(x) \text{ is not differentiable at } x = 0}$$

Problem 8

Trace or copy the graph of the given function f. (Assume that the axes have equal scales.) Then sketch the graph of f' below it.

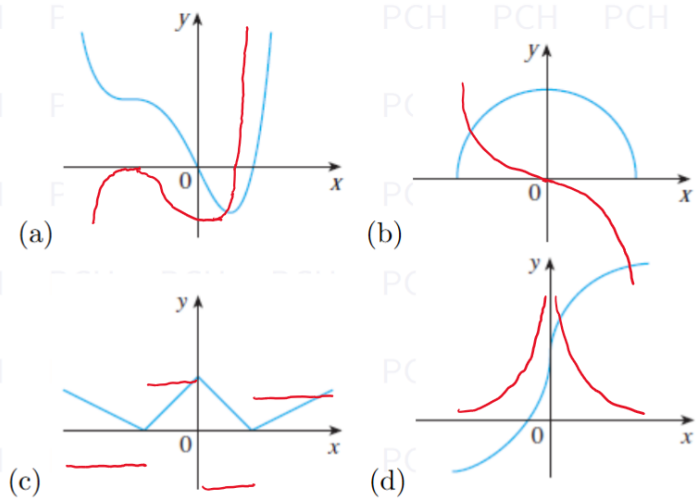
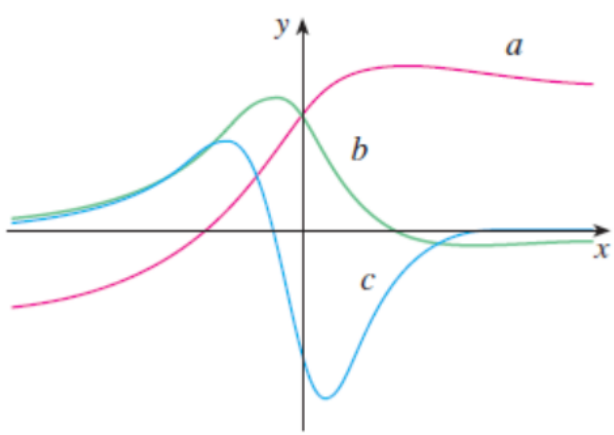


Figure 2.

Problem 9

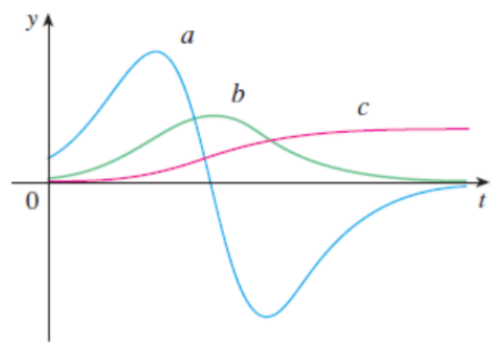
The figure shows the graphs of f, f', and f''. Identify each curve, and explain your choices.



$$a' = 0, b = 0 \implies a' = b$$
$$b' = 0, c = 0 \implies b' = c$$
$$\boxed{f = a, f' = b, f'' = c}$$

Problem 10

The figure shows the graphs of three functions. One is the position function of a car, and one is the velocity of the car, and one is its acceleration. Identify each curve, and explain your choices.



$$c' = 0, b = 0 \implies c' = b$$
$$b' = 0, a = 0 \implies b' = a$$
$$\boxed{f = c, f' = b, f'' = a}$$

Problem 11

Use the limit definition of a derivative to find f' and f''. Then graph f, f', and f'' on a common screen and check to see if your answers are reasonable.

$$f(x) = 1 + 4x - x^2$$

$$\begin{aligned}
f'(x) &= \lim_{x \to h} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{x \to h} \frac{1 + 4(x+h) - (x+h)^2 - 1 - 4x + x^2}{h} \\
&= \lim_{x \to h} \frac{4h - 2hx - h^2}{h} \\
&= \lim_{x \to h} 4 - 2x - h \\
&= \boxed{4 - 2x} \\
f''(x) &= \lim_{x \to h} \frac{f'(x+h) - f'(x)}{h} \\
&= \lim_{x \to h} \frac{4 - 2x - 2h - 4 + 2x}{h} \\
&= \lim_{x \to h} \frac{2h}{h} \\
&= \boxed{2}
\end{aligned}$$

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	−4	4	2
4	−1	3	6	7

Problem 13

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their derivatives at selected values of x. The function h is given by $h(x) = f(g(x)) - 6$. Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

$$h(r) = f(g(r)) - 6 = -5 \implies f(g(r)) = 1$$
$$f(3) = 10, f(4) = -1 \implies 10 < f(g(r)) < -1$$
$$3 < g(r) < 4 \implies 2 < r < 3$$

By the Intermediate Theorem, since $g(x)$ is continuous on $\mathbb{R}$, and $3 = g(2) < g(3) = 4$, then there exist a number $r \in (2, 3)$ such that $g(r) \in (3, 4)$. By the Intermediate Theorem, since $f(x)$ is continuous on $\mathbb{R}$, and $-1 = f(4) < f(3) = 10$, the there exist a number $g(r) \in (3, 4)$ such that $f(g(r)) \in (-1, 10)$. Thus, $h(r)$ is able to achieve the value of -5.

Problem 14

A student starts reading a book at time $t = 0$ minutes and continues reading for the next 10 minutes. The rate at which the student reads is modeled by the differentiable function R, where $R(t)$ is measured in words per minute. Selected values of $R(t)$ (minutes) are given in the table shown.

t (minutes)	0	2	8	10
$R(t)$ (words per minute)	90	100	150	162

(a) Approximate $R'(1)$ using the average rate of change of R over the interval $0 \le t \le 2$. Show the work that leads to your answer. Indicate units of measure.

$$R'(1) = \frac{100 - 90}{2} = \boxed{5 \text{ words/min}^2}$$

(b) Must there be a value c, for $0 < c < 10$, such that $R(c) = 155$? Justify your answer.

1. $R(t)$ is a differentiable and continuous function on [0,10].
2. $R(0) = 90, R(10) = 162$

$$90 < N < 162 \implies \text{let } N = 155$$

3. By the Intermediate Value Theorem, since $R(x)$ is continuous on $[0, 10]$, and $R(0) < 155 < R(10)$, then there exist a number $c \in (0, 10)$ such that $R(c) = 155$.